Larkhall NDE Services Inc.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
LNDE-RPT-20250811-029-R0	Issued	2025-08-15

UT and PAUT Re-measurement - 4-RA-101

Scope and Responsibilities

This report details the 6-month re-measurement of 4-RA-101 within the ATU-100 at the Hamilton Plant. The inspection was executed to monitor localized metal loss downstream of the let-down valve, as identified in CIE-FFS-2025-004-R0. The scope includes ultrasonic thickness measurement and Phased Array Ultrasonic Testing (PAUT) to validate the current corrosion rate against previous assessment assumptions.

Marcus Chen served as the lead UT technician responsible for data acquisition and field analysis. David Park performed the technical review and quality oversight of the NDE data. All activities were conducted in accordance with the requirements specified in QSCP-IP-ATU100 Rev 4 and the site-specific NDE guidelines.

Inspection Methodology and Calibration

Ultrasonic thickness measurements were performed under QSCP-NDT-001 Rev 4, utilizing an Olympus 38DL PLUS 2847193. The gauge was verified against LNDE-CAL-2025-0812. The inspection adhered to the technical requirements of ASTM E797 and API 570 to ensure data repeatability and accuracy for rich amine service piping.

PAUT corrosion mapping was conducted using an Olympus OmniScan X3 1022157, calibrated per LNDE-CAL-2024-0921. This high-resolution mapping was required to evaluate the rich-amine fittings at CML-06, following the validation protocols for CML-id data export and technician certification standards accepted under the current site procedure.

NDE Results and Data Comparison

The primary focus of this re-measurement was the elbow at CML-06, where flashing downstream of the let-down valve has caused localized attack. Current readings were compared against the baseline established in LNDE-RPT-20241202-036-R0. The 4-RA-101 line, constructed of A106 Gr.B with a nominal thickness of 8.56 mm, is subject to a design pressure of 862 kPa(g) and a design temperature of 82 °C.

Data analysis indicates that the corrosion rate remains within the assumptions defined in CIE-FFS-2025-004-R0. While localized thinning is present, the measured degradation does not exceed the limits projected in the Level 2 Fitness-for-Service assessment. The findings support the continued operation of the line under the current monitoring frequency.

Asset Integrity Assessment

At CML-06, a minimum wall thickness of 3.3 mm was recorded. Per site rules, the retirement thickness for 4-RA-101 is the t-min of 3.81 mm. Because the current reading of 3.3 mm is below the t-min, the remaining life is considered expired. This condition necessitates the issuance of an NCR as per the Hamilton Plant asset integrity requirements.

The QSCP-IP-ATU100 Rev 4 inspection plan requires 4-RA-101 to be re-measured in August 2025 and replaced during the August 2026 turnaround. Although the corrosion rate is stable, the thickness at CML-06 confirms the necessity of the planned replacement. All rich amine lines in this class are subject to a 3-year RBI interval under the corporate standard, but the localized flashing at this specific fitting dictates the accelerated 6-month re-measurement cycle.

Thickness readings, measured 2025-08-15

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2024-11-04	6.50	6.32	0.075	0.166	15.1	2028-08-15	
4-RA-101-02	8.56	3.81	2024-11-04	6.46	6.40	0.073	0.170	15.2	2028-08-15	
4-RA-101-03	8.56	3.81	2024-11-04	5.52	5.42	0.108	0.209	7.7	2028-08-15	gauge photo attached
4-RA-101-04	8.56	3.81	2024-11-04	6.90	6.82	0.061	0.124	24.4	2028-08-15	
4-RA-101-05	8.56	3.81	2024-11-04	6.37	6.20	0.082	0.168	14.2	2028-08-15	
4-RA-101-06	8.56	3.81	2024-12-10	3.55	3.30	0.423	0.625	expired	2026-02-13	< t-min; gauge photo attached
4-RA-101-07	8.56	3.81	2024-11-04	7.21	7.10	0.051	0.100	33.0	2028-08-15	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-08	8.56	3.81	2024-11-04	5.06	4.85	0.130	0.275	3.8	2027-07-07	gauge photo attached
4-RA-101-09	8.56	3.81	2024-11-04	7.17	7.04	0.052	0.110	29.4	2028-08-15	
4-RA-101-10	8.56	3.81	2024-11-04	6.91	6.86	0.061	0.118	25.8	2028-08-15	
4-RA-101-11	8.56	3.81	2024-11-04	5.53	5.34	0.113	0.227	6.8	2028-08-15	gauge photo attached
4-RA-101-12	8.56	3.81	2024-11-04	6.10	5.95	0.185	0.185	11.5	2028-08-15	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT/VT Level II; CAN/CGSB-48.9712 UT Level II	2025-08-15	_____
Reviewed	David Park	Level III / Quality Manager, SNT-TC-1A UT Level III	2025-08-15	_____